
E-Commerce Revenue Optimisation & Business Performance Analysis

Business Analytics & Exploratory Data Analysis Project

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PROJECT OVERVIEW

The e-commerce sector generates large volumes of transactional data related to customer purchases, product pricing, order quantities, and geographic markets. However, businesses often fail to fully utilize this data for improving revenue performance and strategic decision-making.

This project analyses an online retail transactional dataset to identify key revenue drivers, evaluate product performance, analyse customer purchasing behaviour, and examine geographic sales patterns.

The analysis aims to generate insights that support **revenue optimisation** and **data-driven business decision-making** in e-commerce environments.

PROJECT OBJECTIVES

The objectives of this project are:

- To analyse **revenue trends** in the e-commerce dataset.
 - To identify **top revenue-generating products**.
 - To evaluate **country-wise sales distribution**.
 - To analyse **customer contribution to revenue**.
 - To derive insights that support **revenue optimisation strategies**.
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TOOLS & TECHNOLOGIES

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Google Colab
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NOTEBOOK WORKFLOW

1. Import Libraries
 2. Load Dataset
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 5. Data Cleaning & Preprocessing
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 11. Country-wise Revenue Analysis
 12. Customer Contribution Analysis
 13. Average Order Value (AOV) Analysis
 14. Revenue Concentration Analysis (Pareto Analysis)
 15. Key Business Insights
 16. Revenue Optimisation Recommendations
 17. Conclusion
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DATASET SOURCE

Online Retail E-Commerce DatasetSource (Kaggle): <https://www.kaggle.com/datasets/carrie1/ecommerce-data>

1. IMPORTING LIBRARIES

```
# IMPORTING LIBRARIES FOR DATA ANALYSIS & VISUALIZATION
import pandas as pd      # for data manipulation
import numpy as np       # for numerical operations

import matplotlib.pyplot as plt  # for plotting charts
import seaborn as sns           # for advanced visualizations

# SETTING CHART STYLE
sns.set(style="whitegrid")

print("Libraries imported successfully")
```

Libraries imported successfully

2. LOADING DATASET

```
# LOADING THE E-COMMERCE DATASET
df = pd.read_csv("OnlineRetail_Ecommerce_Dataset.csv", encoding="ISO-8859-1")

# DISPLAYING FIRST 5 ROWS
df.head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/1/2010 8:26	2.55	17850.0	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	12/1/2010 8:26	3.39	17850.0	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/1/2010 8:26	2.75	17850.0	United Kingdom

3. DATASET OVERVIEW

```
# CHECKING DATASET DIMENSIONS
df.shape
```

(541909, 8)

```
# DISPLAYING COLUMN NAMES
df.columns
```

```
Index(['InvoiceNo', 'StockCode', 'Description', 'Quantity', 'InvoiceDate',
       'UnitPrice', 'CustomerID', 'Country'],
      dtype='object')
```

```
# CHECKING DATA TYPES & MISSING VALUES
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 541909 entries, 0 to 541908
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  -
0   InvoiceNo        541909 non-null object
1   StockCode       541909 non-null object
2   Description     540455 non-null object
3   Quantity        541909 non-null int64
4   InvoiceDate     541909 non-null object
5   UnitPrice       541909 non-null float64
6   CustomerID      406829 non-null float64
7   Country         541909 non-null object
dtypes: float64(2), int64(1), object(5)
memory usage: 33.1+ MB
```

DATASET OVERVIEW – INSIGHTS

Key Observations:

- The dataset contains **541,909 transaction records and 8 variables**, indicating a large retail dataset suitable for business analytics.
 - Each record represents a **product purchased within an invoice transaction**.
 - Important variables such as **Quantity, UnitPrice, CustomerID, and Country** enable analysis of revenue generation, customer behaviour, and geographic sales patterns.
 - The dataset includes transactions from **38 countries**, allowing international market analysis.
-

4. DATASET SUMMARY

```
# BASIC DATASET SUMMARY
```

```
print("Number of rows:", df.shape[0])
print("Number of columns:", df.shape[1])
print("Unique customers:", df['CustomerID'].nunique())
print("Unique products:", df['Description'].nunique())
print("Number of countries:", df['Country'].nunique())
```

```
Number of rows: 541909
Number of columns: 8
Unique customers: 4372
Unique products: 4223
Number of countries: 38
```

DATASET SUMMARY – INSIGHTS

Key Observations:

- The dataset includes **4,372 unique customers**, indicating a broad and diverse customer base.
 - A total of **4,223 unique products** are recorded, reflecting high product variety within the retail catalogue.
 - Transactions originate from **38 countries**, showing the international presence of the business.
 - The large number of customers and products allows meaningful analysis of **purchasing behaviour and key revenue drivers**.
-

5. DATA CLEANING

```
# REMOVING CANCELLED ORDERS (INVOICE STARTING WITH 'C')
df = df[~df['InvoiceNo'].astype(str).str.startswith('C')]
```

```
# REMOVING NEGATIVE QUANTITIES
df = df[df['Quantity'] > 0]
```

```
# REMOVING MISSING CUSTOMER IDS
df = df.dropna(subset=['CustomerID'])
```

```
# CONVERTING DATA COLUMN TO DATETIME
df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])
```

DATA CLEANING – INSIGHTS

Key Steps Performed:

- **Cancelled invoices** were removed to ensure that only completed transactions are included in the analysis.
 - Records with **negative quantities** were filtered out to eliminate product returns and invalid entries.
 - Rows with **missing CustomerID values** were removed to enable accurate customer-level analysis.
 - The **InvoiceDate column was converted to datetime format** to support time-based analysis such as monthly revenue trends.
-

6. FEATURE ENGINEERING

```
# CREATING REVENUE COLUMN
df['Revenue'] = df['Quantity'] * df['UnitPrice']

df.head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Revenue
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850.0	United Kingdom	15.30
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850.0	United Kingdom	22.00

REVENUE VARIABLE

Key Observation:

- A new variable **Revenue** was created by multiplying **Quantity × UnitPrice**.
- This variable represents the **total revenue generated per transaction**.
- The **Revenue column** will be used as the primary metric for analysing sales performance throughout the project.

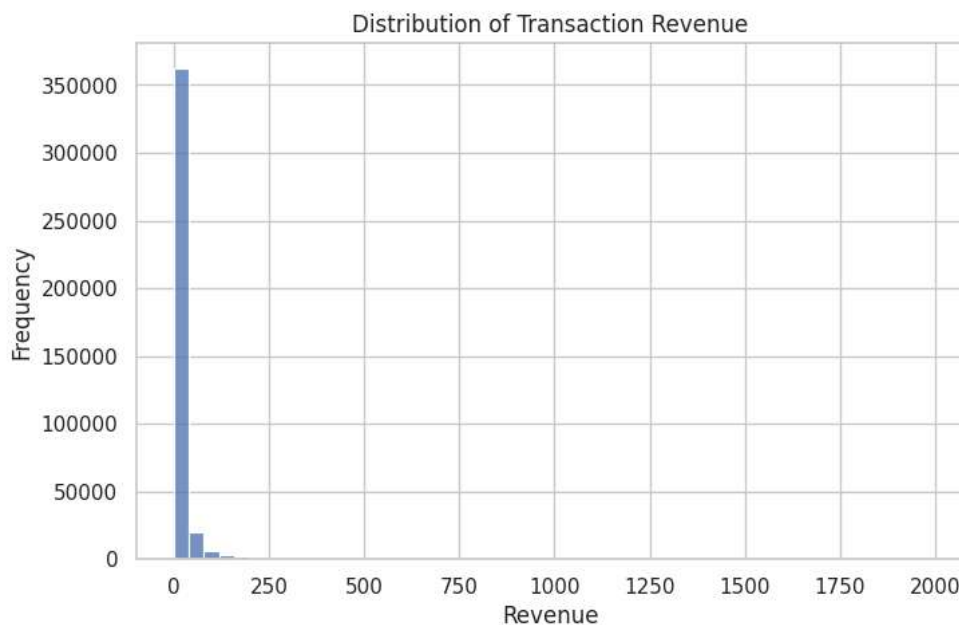
7. REVENUE DISTRIBUTION ANALYSIS

```
# FILTERING EXTREME REVENUE OUTLIERS TO BETTER VISUALIZE THE DISTRIBUTION
revenue_filtered = df[df['Revenue'] < 2000]

plt.figure(figsize=(8,5))
sns.histplot(revenue_filtered['Revenue'], bins=50)

plt.title("Distribution of Transaction Revenue")
plt.xlabel("Revenue")
plt.ylabel("Frequency")

plt.show()
```



REVENUE DISTRIBUTION – INSIGHTS

Key Observations:

- Most transactions generate **low revenue values (largely below 100)**, while only a small number of transactions generate significantly higher revenue.
- This creates a **right-skewed distribution**, where the majority of observations are concentrated at lower revenue levels.

- The distribution indicates that the business experiences **many small purchases alongside relatively few high-value transactions**, which is typical in retail sales data.

8. MONTHLY REVENUE TREND ANALYSIS

```
# EXTRACTING MONTH
df['Month'] = df['InvoiceDate'].dt.to_period('M')

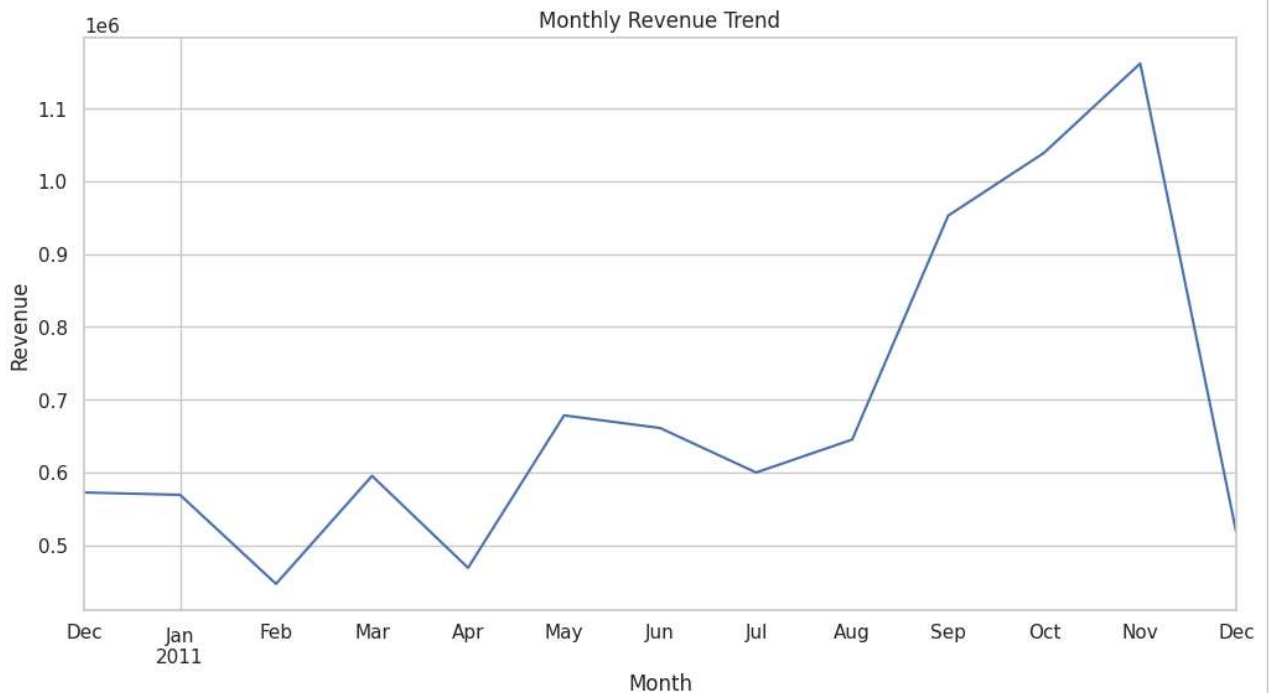
monthly_revenue = df.groupby('Month')['Revenue'].sum()

plt.figure(figsize=(12,6))

monthly_revenue.plot()

plt.title("Monthly Revenue Trend")
plt.xlabel("Month")
plt.ylabel("Revenue")

plt.show()
```



MONTHLY REVENUE TREND – INSIGHTS

Key Observations:

- Monthly revenue fluctuates throughout the year, ranging approximately between **450,000 and 1,160,000**.
- Revenue begins to increase from **September onward**, indicating stronger sales in the later months of the year.
- The **highest revenue occurs in November**, suggesting peak seasonal demand during the holiday shopping period.
- The decline observed in **December** is likely due to **partial data availability**, as the dataset does not include the full month.

9. TOP REVENUE GENERATING PRODUCTS

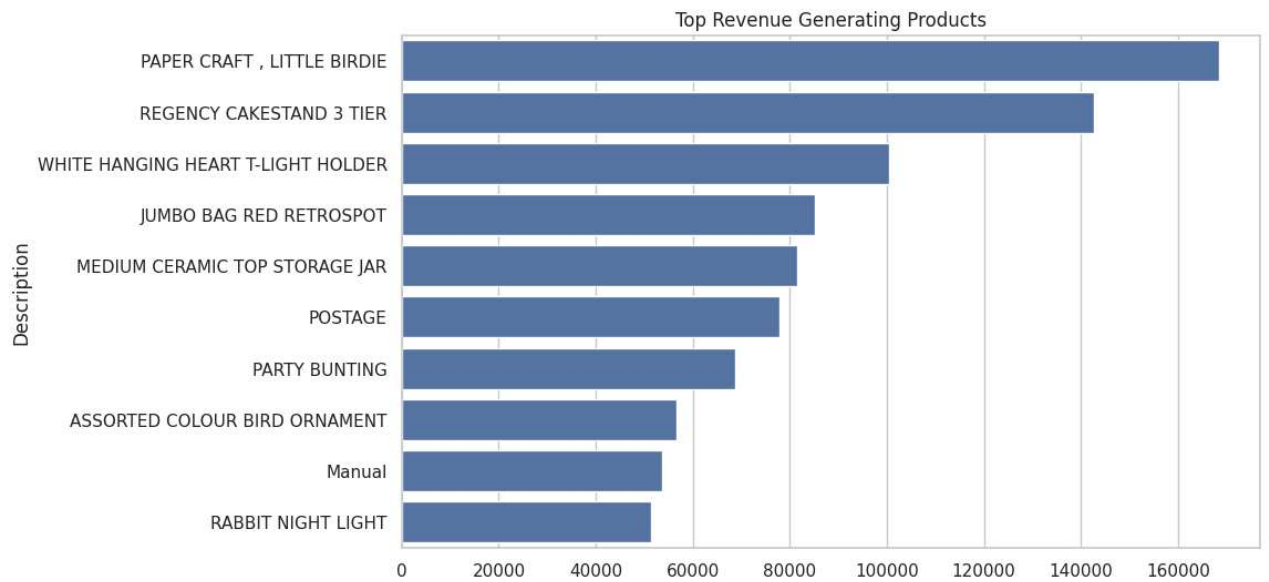
```
# TOP PRODUCTS BY REVENUE
top_products = df.groupby('Description')['Revenue'].sum().sort_values(ascending=False).head(10)

plt.figure(figsize=(10,6))

sns.barplot(x=top_products.values, y=top_products.index)

plt.title("Top Revenue Generating Products")
```

```
plt.show()
```



TOP REVENUE GENERATING PRODUCTS – INSIGHTS

Key Observations:

- **PAPER CRAFT , LITTLE BIRDIE** generates the highest revenue, exceeding **160,000**, making it the top-performing product.
- **REGENCY CAKESTAND 3 TIER** and **WHITE HANGING HEART T-LIGHT HOLDER** also contribute significantly to overall revenue.
- The top 10 products generate revenue ranging approximately from **50,000 to 168,000**.
- This indicates that a **small group of products plays a major role in driving total sales revenue**.

10. TOP PRODUCTS BY QUANTITY SOLD

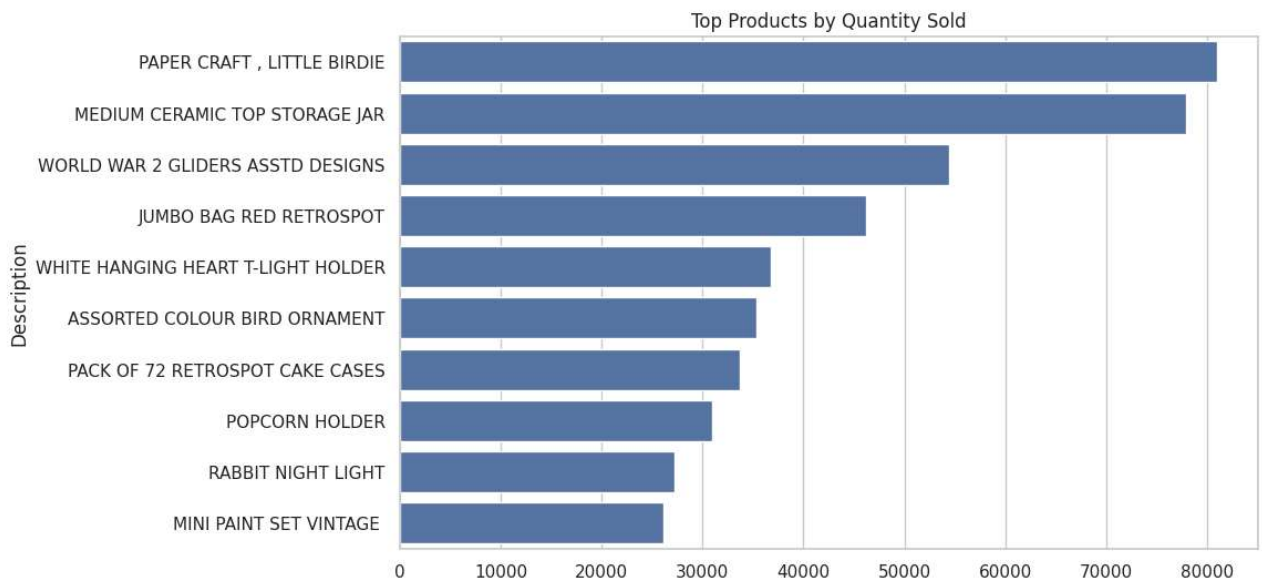
```
# IDENTIFYING PRODUCTS SOLD IN HIGHEST QUANTITIES
top_quantity = df.groupby('Description')['Quantity'].sum().sort_values(ascending=False).head(10)

plt.figure(figsize=(10,6))

sns.barplot(x=top_quantity.values, y=top_quantity.index)

plt.title("Top Products by Quantity Sold")

plt.show()
```



TOP PRODUCTS BY QUANTITY SOLD – INSIGHTS

Key Observations:

- **PAPER CRAFT , LITTLE BIRDIE** has the highest sales volume, exceeding **80,000 units sold**.
- **MEDIUM CERAMIC TOP STORAGE JAR** and **WORLD WAR 2 GLIDERS ASSTD DESIGNS** also show strong demand.
- The top products have sales volumes ranging approximately from **25,000 to 80,000 units**.
- This indicates that certain products experience **very high customer demand and frequent purchases**.

11. COUNTRY-WISE REVENUE ANALYSIS

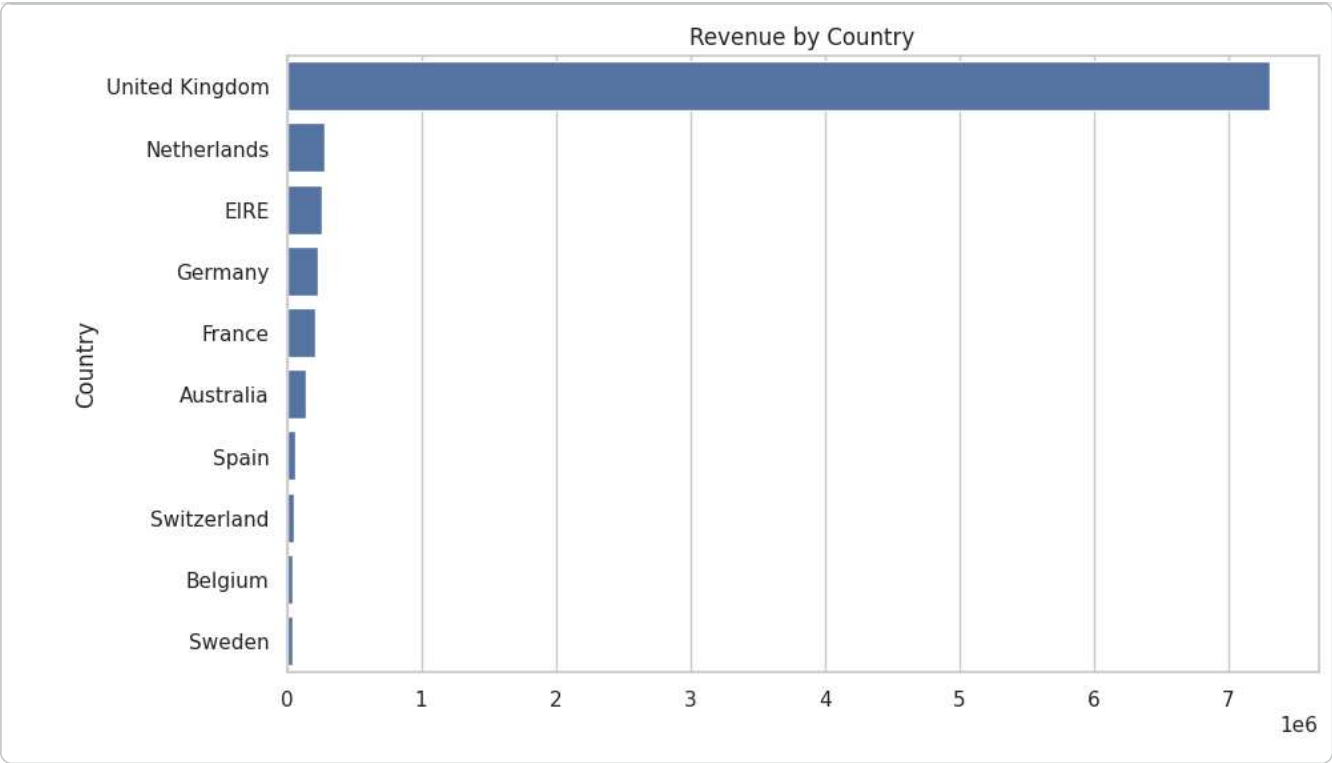
```
# REVENUE BY COUNTRY
country_revenue = df.groupby('Country')['Revenue'].sum().sort_values(ascending=False).head(10)

plt.figure(figsize=(10,6))

sns.barplot(x=country_revenue.values, y=country_revenue.index)

plt.title("Revenue by Country")

plt.show()
```



COUNTRY-WISE REVENUE ANALYSIS – INSIGHTS

Key Observations:

- The **United Kingdom generates the highest revenue**, exceeding **7 million**, making it the dominant market.
- Other countries such as **Netherlands, EIRE, Germany and France** contribute significantly smaller portions of revenue.
- The large gap between the UK and other countries indicates **strong geographic concentration of sales**.
- This suggests that the business is **highly dependent on the UK market**, with international markets contributing relatively smaller shares.

12. CUSTOMER CONTRIBUTION ANALYSIS

```
# CALCULATING TOTAL REVENUE GENERATED BY EACH CUSTOMER
customer_revenue = df.groupby('CustomerID')['Revenue'].sum().sort_values(ascending=False).head(10)

customer_revenue
```

Revenue	
CustomerID	
14646.0	280206.02
18102.0	259657.30
17450.0	194550.79
16446.0	168472.50
14911.0	143825.06
12415.0	124914.53
14156.0	117379.63
17511.0	91062.38
16029.0	81024.84
12346.0	77183.60

dtype: float64

```
# CONVERTING CUSTOMERID FROM FLOAT TO INTEGER FOR CLEANER LABELS
customer_revenue.index = customer_revenue.index.astype(int)

# PLOTTING TOP CUSTOMERS BY REVENUE
```



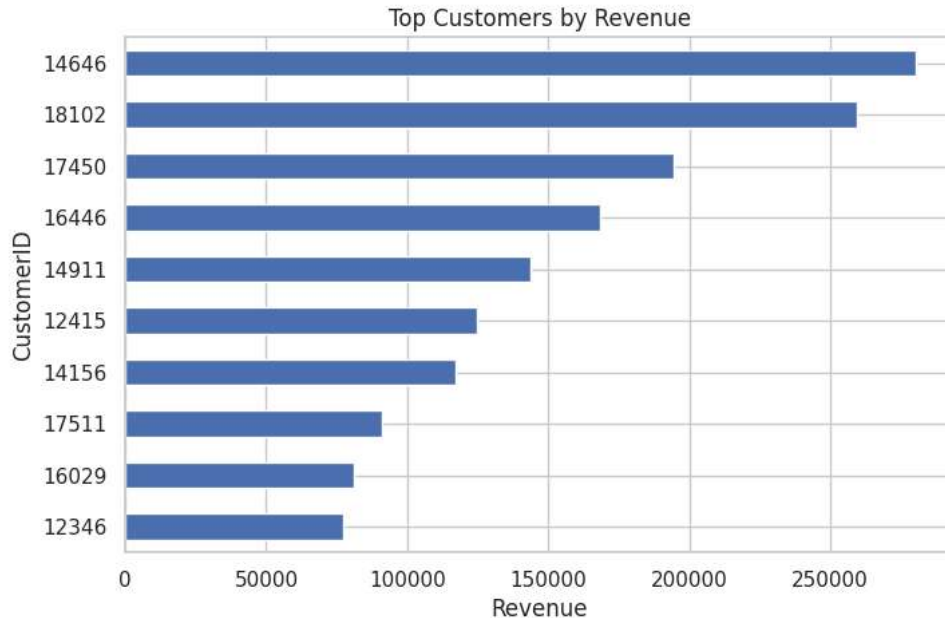
```
plt.figure(figsize=(8,5))

customer_revenue.plot(kind='barh')

plt.title("Top Customers by Revenue")
plt.xlabel("Revenue")
plt.ylabel("CustomerID")

# INVERTING Y-AXIS SO HIGHEST VALUE APPEARS AT TOP
plt.gca().invert_yaxis()

plt.show()
```



CUSTOMER REVENUE CONTRIBUTION – INSIGHTS

Key Observations:

- Customer **14646** generates the highest revenue, exceeding **270,000**.
- Customers **18102** and **17450** also contribute significantly to total sales.
- The top 10 customers generate revenue ranging approximately from **75,000 to 280,000**.
- This indicates that a **small group of high-value customers contributes a substantial share of total revenue**, highlighting the importance of customer retention strategies.

13. AVERAGE ORDER VALUE (AOV) ANALYSIS

```
# CALCULATING AVERAGE ORDER VALUES
total_revenue = df['Revenue'].sum()
total_orders = df['InvoiceNo'].nunique()

AOV = total_revenue / total_orders

print("Average Order Value:", round(AOV,2))
```

Average Order Value: 480.76

AVERAGE ORDER VALUE (AOV) – INSIGHTS

Key Observations:

- The **Average Order Value (AOV)** is approximately **480.76**, meaning that each order generates around **480 in revenue on average**.
- A higher AOV suggests that customers often purchase **multiple items or higher-value products within a single transaction**.
- Businesses can increase AOV through strategies such as **product bundling, cross-selling, and targeted promotional offers**.

14. REVENUE CONCENTRATION (PARETO ANALYSIS)

```
# ANALYZING REVENUE CONCENTRATION
product_revenue = df.groupby('Description')['Revenue'].sum().sort_values(ascending=False)

top_20_percent = product_revenue.head(int(len(product_revenue)*0.2))

print("Revenue from top 20% products:", round(top_20_percent.sum(),2))

Revenue from top 20% products: 7036402.86
```

REVENUE CONCENTRATION (PARETO ANALYSIS) – INSIGHTS

Key Observations:

- The **top 20% of products** generate approximately **7,036,402.86** in total revenue, indicating strong revenue concentration.
- This shows that a **small proportion of products** contributes a **significant share of overall sales**.
- The pattern reflects the **Pareto Principle (80/20 rule)**, where a limited number of products drive a large portion of total revenue in retail businesses.

15. KEY BUSINESS INSIGHTS

Key Observations:

- Revenue is **concentrated among a limited number of high-performing products**, indicating strong product-level revenue concentration.
- The **United Kingdom** contributes the **largest share of total revenue**, highlighting strong geographic dependence on a single market.
- Monthly revenue analysis reveals **seasonal fluctuations in sales**, with higher revenue observed toward the end of the year.
- A **small group of high-value customers** generates a **significant portion of total revenue**, emphasizing the importance of customer retention.
- Effective **product portfolio management and focus on high-performing items** can significantly support revenue optimization.

16. REVENUE OPTIMIZATION RECOMMENDATIONS

Strategic Recommendations:

- Focus **marketing and promotional efforts on top revenue-generating products** identified in the analysis.
- Develop **loyalty programs and targeted offers** for high-value customers who contribute significantly to overall revenue.
- Increase **Average Order Value (AOV ≈ 480.76)** through product bundling, cross-selling, and targeted promotional campaigns.
- Expand **international sales markets** to reduce strong dependence on the United Kingdom market.
- Improve **inventory planning and stock management** by aligning supply with observed seasonal demand patterns.

17. CONCLUSION

This project analysed a large e-commerce transactional dataset containing over **541,000 records** to identify key revenue drivers and business performance patterns.

The analysis revealed that revenue is **concentrated among a limited number of products and customers**, consistent with the **Pareto Principle (80/20 rule)**. The **United Kingdom** contributes the **majority of total revenue**, indicating strong geographic concentration.

Seasonal sales patterns were also identified, with revenue increasing toward the **later months of the year**, while the **Average Order Value (AOV) was approximately 480.76 per order**.

Overall, the analysis demonstrates how **data analytics can help businesses understand customer behaviour, identify revenue opportunities, and support data-driven decision-making in the e-commerce sector**.

Double-click (or enter) to edit